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Maximum-Likelihood Estimation

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EEE 554 Lecture #14

Given a family $\{f_X(x|\theta)\}$ of probability densities parameterized by $\theta \in \mathbb{R}^n$ and data $X_1, \dots, X_n$ drawn from one of these PDFs, the goal of estimation is to determine a value for θ that is most compatible with the data in some sense.

One such sense is **maximum likelihood**.

Suppose $\underline{X} = (X_1, \dots, X_n)^T$ with each X_k having PDF $f_X(x|\theta)$. The likelihood function is:

$$l(\theta, \underline{x}) = f_{\underline{X}}(\underline{x}|\theta)$$

The log-likelihood function is:

$$L(\theta, \underline{x}) = \log l(\theta, \underline{x}) = \log(f_{\underline{X}}(\underline{x}|\theta))$$

Note that log is monotonic, so maximizing $L(\theta, \underline{x})$ is equivalent to maximizing $l(\theta, \underline{x})$.

A maximum-likelihood (ML) estimator for θ from data $\underline{x}$ has the form:

$$\hat{\theta}_{ML} = \arg \max_{\theta} l(\theta, \underline{x}) = \arg \max_{\theta} L(\theta, \underline{x})$$

Note that $\hat{\theta}_{ML}$ may not be unique because it is possible that more than one value of θ maximizes $l(\theta, \underline{x})$.

Example: Suppose $X_1, \dots, X_n$ are iid normal RVs with known variance σ^2 and unknown mean $\theta = \mu$.

So,

$$\begin{aligned} l(\mu, \underline{x}) &= f_{\underline{X}}(\underline{x}|\mu) = f_X(x_1|\mu) \dots f_X(x_n|\mu) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x_1 - \mu)^2}{2\sigma^2} \right\} \right) \dots \left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x_n - \mu)^2}{2\sigma^2} \right\} \right) \\ &= \frac{1}{(2\pi\sigma^2)^{n/2}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{k=1}^n (x_k - \mu)^2 \right\} \end{aligned}$$

Taking the log:

$$L(\mu, \underline{x}) = -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{k=1}^n (x_k^2 - 2\mu x_k + \mu^2)$$

Taking the derivative:

$$\frac{d}{d\mu} L(\mu, \underline{x}) = \frac{1}{2\sigma^2} \sum_{k=1}^n (2x_k - 2\mu)$$

Setting this to 0:

$$n\mu - \sum_{k=1}^n x_k = 0$$

Thus, $\hat{\mu}_{ML} = \frac{1}{n} \sum_{k=1}^n x_k$

Example: Now suppose $X_1, \dots, X_n$ are iid normal RVs with known mean μ and unknown variance $\theta = \sigma^2$.

So,

$$L(\theta, \underline{x}) = L(\sigma^2, \underline{x}) = -\frac{n}{2} \log(2\pi\theta) - \frac{1}{2\theta} \sum_{k=1}^n (x_k - \mu)^2$$

Taking the derivative w.r.t θ :

$$\begin{aligned} \frac{d}{d\theta} L(\theta, \underline{x}) &= -\frac{n}{2} \frac{1}{2\pi\theta} (2\pi) + \frac{1}{2\theta^2} \sum_{k=1}^n (x_k - \mu)^2 \\ &= -\frac{n}{2\theta} + \frac{1}{2\theta^2} \sum_{k=1}^n (x_k - \mu)^2 \end{aligned}$$

Assuming $\theta \neq 0$, setting $\frac{d}{d\theta} L(\theta, \underline{x}) = 0 \Rightarrow \hat{\theta}_{ML} = \frac{1}{n} \sum_{k=1}^n (x_k - \mu)^2$

Example (Tale of two estimators): Take $X_1, \dots, X_n$ to be iid $\sim U[0, b]$. The parameter to be estimated is $\theta = b$.

$$l(b, \underline{x}) = f_{\underline{X}}(\underline{x}|b) = \begin{cases} \left(\frac{1}{b}\right)^n & 0 \leq x_k \leq b \text{ for all } k \\ 0 & \text{else} \end{cases}$$

This is maximized by taking b to be as small as possible while still satisfying $x_k \leq b$ for all $k = 1, \dots, n$; i.e.,

$$\hat{\theta}_{ML} = \max(x_1, \dots, x_n)$$

The CDF of $\hat{\theta}_{ML}$ is:

$$F_{\hat{\theta}}(v) = P(\{X_1 \leq v\} \cap \dots \cap \{X_n \leq v\}) = (F_X(v))^n$$

Thus,

$$f_{\hat{\theta}}(v) = n(F_X(v))^{n-1} f_X(v) = n \left(\frac{v}{b}\right)^{n-1} \left(\frac{1}{b}\right) = n \frac{v^{n-1}}{b^n}$$

And,

$$E[\hat{\theta}_{ML}] = \int_0^b v \left(n \frac{v^{n-1}}{b^n}\right) dv = \frac{n}{b^n} \int_0^b v^n dv = \frac{n}{n+1} b$$

So, $\hat{\theta}_{ML}$ is biased with:

$$\beta(\hat{\theta}_{ML}) = E[\hat{\theta}_{ML}] - b = \left(\frac{n}{n+1} - 1\right) b = -\frac{1}{n+1} b$$

Note that $\beta \rightarrow 0$ as $n \rightarrow \infty$, so it is asymptotically unbiased.

To get $\text{var}(\hat{\theta}_{ML})$, consider:

$$E[\hat{\theta}_{ML}^2] = \frac{n}{b^n} \int_0^b v^{n+1} dv = \frac{n}{n+2} b^2$$

$$\text{var}(\hat{\theta}_{ML}) = E[\hat{\theta}_{ML}^2] - (E[\hat{\theta}_{ML}])^2 = \frac{n}{n+2} b^2 - \left(\frac{n}{n+1} b\right)^2 = \frac{n}{(n+1)^2(n+2)} b^2$$

Compare this to the linear estimator $\tilde{\theta} = \frac{2}{n} \sum_{k=1}^n X_k$ for which:

$$E[\tilde{\theta}] = \frac{2}{n} \cdot n \left(\frac{b}{2} \right) = b$$

This estimator of b is unbiased. Its variance is four times that of the sample mean estimator; i.e.,

$$\text{var}(\tilde{\theta}) = 4 \frac{\text{var}(X)}{n} = 4 \frac{b^2/12}{n} = \frac{1}{3n} b^2$$

While biased, $\hat{\theta}_{ML}$ has a variance that decreases quadratically with n ($\sim 1/n^2$). The variance of the unbiased linear estimator $\tilde{\theta}$ decreases only linearly with n ($\sim 1/n$).